

Weighted Schur at $L = 0.3$:
 A -GO, finite S_w PD, operator M_∞ open
— 0.3 keystone checkpoint —
PRIME.RDAGGER.WEIGHTED_SCHUR.01 — round 485

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Abstract

Rounds 479–484 used the *flat* tail floor $c_{\text{tail}} = \sigma_A(8) \approx 0.2409$. But $\sigma_A(t)$ grows like $\log(t/2\pi)$. The leftover coupling $\|C_{\text{rest}}\| \approx 0.294$ pairs the certified block with the orthocomplement, whose A -reserve is frequency-weighted. The reviewer path-B test is $\|M_\infty^{-1/2} C M_{00}^{-1/2}\| \leq 1$, i.e. the Schur subtract $CM_\infty^{-1}C^\top$ rather than $CC^\top/c_{\text{tail}}$. **A -first (hard Go/No-Go).** On the resolved $n = 96$ surface the leftover C of $\{\psi_0, \dots, \psi_8\}$ into $\{\psi_{10}, \dots, \psi_{22}\}$ has Hilbert–Schmidt mass $[0, 8]$: $2.74 \cdot 10^{-10}$ (the ε -control: almost no band mass); $[8, 30]$: $2.97 \cdot 10^{-2}$; $[30, 150]$: $3.10 \cdot 10^{-1}$; hi-frac = 0.991. **GO:** the 0.294-mass sits at large t , not near $t_* = 8$. The top complement singular vector has $Q_A[0, 150]/c_{\text{tail}} = 4.49$. **Finite weighted Schur.** $S_w = B - CD^{-1}C^\top$ on the $n = 96$ even compression is PD: at $T = 150$, $\lambda_{\min}(S_w) = 6.38 \cdot 10^{-3}$ (Higham $\mu \geq 2.59 \cdot 10^{-3}$), beating $\lambda_{\min}(S_{\text{flat}}) = 5.18 \cdot 10^{-3}$. At $T = 400$, $d_{\min}(D) = 1.49$ and $\lambda_{\min}(S_w) = 7.23 \cdot 10^{-3}$. This is a lemma on V_{96} , not λ_* . The operator $M_\infty = (1 - P)A(1 - P)$ on the true L^2 orthocomplement is not enclosed: a higher uniform floor $\sigma(t_1)$ does not follow from ε of K_8 , and the L^2 leftover of V_{96} is the same completeness hole as r483/r484. Verdict REDUCED(A -GO; finite- S_w -PD; operator- M_{inf} -open)+CHECKPOINT. Fourier–HS is not reused. Yoshida–Bombieri zone. No RH claim.

Status. REDUCED(A -GO; finite- S_w -PD; operator- M_{inf} -open)+CHECKPOINT. No promotion of an RH claim. This is the last structurally new 0.3 idea (the block-extension hypothesis fell twice). The 0.3 keystone is frozen here.

Claim boundary. Nothing here is $\lambda_*(0.3) \geq c$. The A -measurement is GO; the finite weighted Schur is PD; the operator M_∞ on the true complement is open. No L^* claim, no $R^\dagger$ claim, no RH claim, no anti-RH claim.

1 Convention

$\text{supp } h \subset [-L, L]$, $g = h * \tilde{h}$, $Q_W = A - P + \Pi$, prime nodes $n \leq e^{2L}$. At $L = 0.3$ one has $P \equiv 0$ (scramble-sensitive, vacuous). A is the Fourier multiplier with symbol $\sigma_A(t) = \text{Re } \psi(\frac{1}{4} + it/2) - \log \pi$ (even, increasing on $[0, \infty)$). $\hat{h}(t) = \int h e^{itx} dx$, $\|h\|_2^2 = (1/2\pi) \int |\hat{h}|^2$, $\langle h, Ah \rangle = (1/2\pi) \int \sigma_A |\hat{h}|^2$. The r469/r479–r484 contract is in force: $t_* = 8$, $\varepsilon = \lambda_6(K_8) \leq 1.46 \cdot 10^{-8}$, $c_{\text{tail}} \geq 0.2408$, Filon $[0, 150]$ and four-term IBP as in r483, certified even block $\{\psi_0, \psi_2, \psi_4, \psi_6, \psi_8\}$.

2 A -first: frequency of leftover C

On $n = 96$, leftover $C_{ij} = \langle \psi_i, A\psi_j \rangle$ for $i \in \{0, 2, 4, 6, 8\}$, $j \in \{10, 12, \dots, 22\}$, split by frequency bands of A :

Band	$\ C\ _F$	$\ C\ _{\text{op}}$	comment
[0, 8]	$2.74 \cdot 10^{-10}$	$2.74 \cdot 10^{-10}$	ε : no band mass
[8, 20]	$4.71 \cdot 10^{-4}$	$4.71 \cdot 10^{-4}$	near t_* , negligible
[8, 30]	$2.97 \cdot 10^{-2}$	—	low tail
[20, 50]	$1.68 \cdot 10^{-1}$	$1.48 \cdot 10^{-1}$	—
[30, 150]	$3.10 \cdot 10^{-1}$	—	high tail
[50, 150]	$3.30 \cdot 10^{-1}$	$2.98 \cdot 10^{-1}$	—
[0, 150] full	$3.05 \cdot 10^{-1}$	$2.94 \cdot 10^{-1}$	matches r484
[150, 400]	$4.23 \cdot 10^{-1}$	$4.19 \cdot 10^{-1}$	more than [0, 150]

Hi-frac $\|C_{[30,150]}\|_F^2 / (\|C_{[8,30]}\|_F^2 + \|C_{[30,150]}\|_F^2) = 0.991$. The top right singular vector of $C_{[0,150]}$ has hat- L^2 energy $[0, 8] \sim 10^{-19}$, $[8, 20] \sim 2 \cdot 10^{-7}$, $[20, 50] = 8.4 \cdot 10^{-3}$, $[50, 150] = 0.37$, $[150, 400] = 0.45$, and $Q_A[0, 150] = 1.08$ versus $c_{\text{tail}} = 0.241$ (ratio 4.49). **GO**. If the mass had sat near $t = 8$ the weighted route would have died in minutes; it does not.

Edge bookkeeping is explicit: ε of K_8 kills band mass of the complement on $[0, 8]$ ($\langle K_8 v, v \rangle \leq \lambda_{10}$ for v in the K_8 -complement of the certified block). It does *not* by itself give a higher floor $\sigma(t_1)$ for $t_1 > 8$.

3 Finite weighted Schur

Write $B = Q|_V$, $C = P_V Q P_W$, $D = Q|_W$ for the even Nyström compression on $n = 96$, $V = \text{span}\{\psi_0, \psi_2, \psi_4\}$ (or the five-mode block), W the even rest in V_{96} . The flat Schur is $S = B - C C^\top / c_{\text{tail}}$; the weighted Schur is $S_w = B - C D^{-1} C^\top$.

Setup	T	$d_{\min}(D)$	$\ C\ $	$\lambda_{\min} S$	$\lambda_{\min} S_w$	Higham μ
B_3 vs rest 6+	150	0.152	0.273	$5.18 \cdot 10^{-3}$	$6.38 \cdot 10^{-3}$	$2.59 \cdot 10^{-3}$
B_5 vs rest 10+	150	0.155	0.294	$5.81 \cdot 10^{-3}$	$6.37 \cdot 10^{-3}$	$2.69 \cdot 10^{-3}$
B_3 vs rest 6+	400	1.49	0.378	$7.08 \cdot 10^{-3}$	$7.23 \cdot 10^{-3}$	$3.31 \cdot 10^{-3}$
odd B_3 vs rest	150	0.177	0.291	0.203	0.221	easy
$n = 48$ r483-style	150	1.22	—	$5.686 \cdot 10^{-3}$	$6.53 \cdot 10^{-3}$	hits r483

Lemma 1 (finite weighted Schur, V_{96}). *On the even $n = 96$ compression at $T = 150$, $\lambda_{\min}(S_w) \geq 2.59 \cdot 10^{-3}$ by Higham with entry radius $2 \cdot 10^{-4}$. The same statement at $T = 400$ holds with $\mu \geq 3.01 \cdot 10^{-3}$. Odd $S_w \geq 0.15$.*

The gain over the flat Schur is modest ($5.2 \cdot 10^{-3} \rightarrow 6.4 \cdot 10^{-3}$), not a $4.5\times$ miracle: at $T = 150$ one has $d_{\min}(D) = 0.152 < c_{\text{tail}}$ because the truncated Gram misses the high- t mass of D (the $[150, 400]$ band of C is larger than $[0, 150]$). At $T = 400$, $d_{\min}(D) = 1.49$ and D is well conditioned. Midpoint $\text{inv}(D)$ is not an interval certificate of the *operator*; Higham is applied to the finite matrix S_w after the discrete inverse.

4 Why this is not λ_*

Three distinct gaps remain.

1. Discrete D is a *compression* of $M_\infty = (1 - P)A(1 - P)$, not the operator. A lower bound on D (so that D^{-1} is controlled from above) on the true L^2 complement is open.
2. ε of K_8 only kills hat-mass on $[0, 8]$. A higher uniform floor $\sigma(t_1)$ for $t_1 > 8$ on the K_8 -complement needs a new argument (Slepians of K_{t_1} , or a Markov bound).
3. The L^2 leftover of the Nyström space V_{96} is the same completeness hole as r483/r484. Certified jets stop at ψ_8 (r484 garbage criterion). There is no certified ONB of the complement on which to run Filon/ 1σ .

The flat majorant $D \succeq c_{\text{tail}}I$ is already false on the $T = 150$ discrete rest ($d_{\min} = 0.15$). The flat Schur that uses c_{tail} can be optimistic on a truncated $[0, T]$ Gram; that is why the $T = 400$ column is recorded.

5 Checkpoint STATE (r479–r485)

What is proved, load-bearing, and not to be regressed:

Round	Verdict	Sealed content
r478	PARTIAL	even Fourier Gram PD; tail-Schur shape
r479	REDUCED	$\kappa_{\text{high}} \leq 1.79 \cdot 10^{-2}$ at $t_* = 8$, $N = 5$ (midpoint)
r480	REDUCED	refined matrix Schur $S = B - CC^\top / c_{\text{tail}}$; crude κ_{high} killed at 0.30
r481	REDUCED	interval FT + IBP tail; $\varepsilon = \lambda_6(K_8) \leq 1.46 \cdot 10^{-8}$, $c_{\text{tail}} \geq 0.2408$
r482	STUCK	Carleson/SR; disjoint from λ_*
r483	REDUCED	Filon $[0, 150]$ + 4-term IBP; finite even S Higham $\mu \geq 1.20 \cdot 10^{-3}$; leftover $\ C\ _F = 2.76 \cdot 10^{-2}$ vs budget $2.4 \cdot 10^{-4}$ ($n = 48$)
r484	STUCK	20× drop falsified at $n = 96$; jets $\geq \psi_{10}$ garbage at $n = 128$; QR-IBP kills S ; enlarge- B leftover $0.294 = \kappa_{\text{high}}$
r485	REDUCED+CHECKPOINT	A-GO (hi-frac 0.991); finite S_w interval-PD; operator M_∞ open

Proved machinery (keep): frequency form of A ; t_* -lemma; $\varepsilon/c_{\text{tail}}$ lifts; Filon and IBP integral classes; finite (flat and weighted) Schur interval-PD on V_n .

Open, one object: the coupling of the certified five-mode even block into its true L^2 orthocomplement, measured against the *weighted* archimedean reserve M_∞ , not against the flat floor $c_{\text{tail}}I$.

6 Open problem (Erdős form)

The remainder is a self-contained analytic question. It does not mention the Riemann hypothesis, primes, or the von Mangoldt comb ($P \equiv 0$ at this L).

Problem 1 (Weighted archimedean coupling). *Bound the coupling of the certified 5-mode even Slepian block into its orthocomplement against the weighted archimedean reserve.*

Data. Let $L = 3/10$. Let A be the (even) Fourier multiplier on $L^2[-L, L]$ with symbol $\sigma_A(t) = \text{Re} \psi(\frac{1}{4} + it/2) - \log \pi$, normalised so $\langle u, Au \rangle = (1/2\pi) \int_{\mathbb{R}} \sigma_A(t) |\hat{u}(t)|^2 dt$. Let $K_{t_*} = \mathbf{1}_{[-t_*, t_*]}(D)$ in the Fourier picture (the time-band-limited projection at $t_* = 8$), and let $\psi_0, \psi_2, \psi_4, \psi_6, \psi_8$ be the first five even eigenfunctions of K_8 on $[-L, L]$ (prolate spheroidal / Slepian; $\lambda_0 \geq \dots \geq \lambda_8$, with $\lambda_6 \leq 1.46 \cdot 10^{-8}$ already certified). Write $V = \text{span}\{\psi_0, \psi_2, \psi_4, \psi_6, \psi_8\}$ and $V^\perp$ its orthogonal complement in $L^2[-L, L]$. Let P_V be the orthogonal projection onto V . Set

$$B = P_V(A + \Pi)P_V, \quad C = P_V(A + \Pi)(1 - P_V), \quad M_\infty = (1 - P_V)(A + \Pi)(1 - P_V),$$

where Π is the even rank-one $\Pi = 2|\langle \cdot, \cosh(x/2) \rangle|^2$.

Ask. Prove either of the following (they are equivalent for the Schur test, up to the already-certified $\lambda_{\min}(B)$ and the ε -bookkeeping on $[0, 8]$):

- (i) $\|M_\infty^{-1/2} C B^{-1/2}\| \leq 1 - \delta$ for an explicit $\delta > 0$ (reviewer path B);
- (ii) the operator Schur complement $S = B - CM_\infty^{-1}C^*$ satisfies $\lambda_{\min}(S) \geq c$ for an explicit $c > 0$ (then $\lambda_*(0.3) \geq c$ on the even sector, and the odd sector is easier).

A useful sufficient form is a quadratic-form lower bound $\langle v, M_\infty v \rangle \geq \int_{|t| \geq 8} \sigma_A(t) |\hat{v}(t)|^2 \frac{dt}{2\pi} - E(v)$ for $v \in V^\perp$, with E controlled by ε (band mass on $[0, 8]$ is $\leq \varepsilon \|v\|_2^2$) plus an explicit time-band edge. The $1/\sigma$ -weighted integrals that would enter a Filon/IBP attack on $CM_\infty^{-1}C^*$ are the same class as r483, only with a $1/\sigma$ factor.

What is already known, and may be used. σ_A is even and increasing on $[0, \infty)$; $\sigma_A(8) \geq 0.2408$; $V^\perp$ has almost no K_8 -mass ($\varepsilon \leq 1.46 \cdot 10^{-8}$); the leftover operator norm of C on the $n = 96$ Nyström surface is 0.294; 99.1% of that leftover HS-mass on $[8, 150]$ sits in $[30, 150]$; the top singular vector of the discrete C has $Q_A[0, 150]/c_{\text{tail}} = 4.49$; the finite compression S_w on V_{96} is interval-PD. Certified jets for IBP stop at ψ_8 ; modes ψ_{10+} are garbage at every $n \leq 128$.

What is forbidden as a pad. The flat floor $M_\infty \succeq c_{\text{tail}}I$ used as if it were the operator (already false on the $T = 150$ discrete rest); Fourier–HS majorants; crude κ_{high} as a SATZ pad; integer Fourier ONB; σ^2 Plancherel as $\|Au\|_{[-L, L]}$; IBP envelope as pad; exponential-sum Ci to ∞ ; the $n = 48$ leftover $2.76 \cdot 10^{-2}$ as a SATZ pad; a $20\times$ coupling drop per even double-step; QR sinc-IBP of garbage jets; any representation of $V^\perp$ by unresolved Nyström nullspace modes.

Honest type. This is an *ideas* problem, not a remaining arithmetic grind. The frequency distribution is measured; the Filon/IBP machine stands; the finite S_w is PD. What is missing is a handle on M_∞ that does not represent $V^\perp$ by garbage eigenmodes: either a genuine lower bound for the weighted quadratic form on the true complement, or a completeness statement that the leftover of V_{96} is smaller than the Higham margin. A higher uniform floor $\sigma(t_1)$ is a new analytic lemma, not a tighter quadrature.

7 Kills

Object	Round	Status
Fourier–HS majorant	r478	not reused
Crude κ_{high}	r480	killed; lives in complement
Integer Fourier ONB	r481	$Q_{0n} \sim 1/n$
σ^2 as $\ Au\ _{[-L, L]}$	r481	killed
IBP envelope as pad	r481	killed
Exp-sum Ci to ∞	r483	killed
$n = 48$ leftover as SATZ pad	r484	undercount
$20\times$ drop / block through ψ_{18}	r484	falsified
High-mode jets $n = 128$	r484	garbage $\geq \psi_{10}$
QR sinc-IBP tail	r484	$\lambda_{\min}(S) < 0$
Flat $c_{\text{tail}}I$ as M_∞	r485	$d_{\min}(D) < c_{\text{tail}}$ at $T = 150$
Weighted Schur as SATZ pad	r485	finite S_w only

8 Scaling at $L = 0.8$

The A -measurement, Filon/IBP, and finite S_w machine survive (P is a cosine sum; scramble becomes live). The operator gap does not become easier: it is a statement about M_∞ on $V^\perp$, not about T or about L . At $L = 0.8$ one must additionally control P on the prime range $n \leq e^{2L}$.

9 What is not claimed

No $\lambda_*(0.3) \geq c$. The A -Go/No-Go is GO; the finite weighted Schur is interval-PD on V_{96} ; the operator M_∞ on the true complement is open. The 0.3 keystone is checkpointed. No L^* claim, no $R^\dagger$ claim, no RH claim, no anti-RH claim.